

A Reduced Economy of Automation, Direct Demand, and Opportunity

A self-contained reproducible example

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Abstract

This document isolates the worked quantitative example that accompanies *Automation, Institutions, and Economic Freedom*. It joins a liquid-income displacement, a fiscal collection rule, household incidence, public allocation, institutional capacity, and a future opportunity service. The exercise has two dates and two household groups. It measures direct expenditure at fixed prices; it is not a solved HANK equilibrium, an empirical calibration, or an estimate. Every displayed table and every plotted coordinate is generated from the machine-readable output of one verified run.

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1 Question, timing, and scope

The example asks when a common automation shock and public budget can pass two different gates:

$$\Delta D_0^{\text{dir}} \geq 0, \quad \text{date-1 opportunity} \geq \mathcal{O}_{\min}.$$

The first gate records immediate incidence and first-round expenditure. The second requires both installed capacity and a funded operating input. Separating the gates makes it possible for a policy to pass one, both, or neither.

There is a unit mass of households. Group H is exposed to the automation income shift and group S owns the corresponding claim. Group variables are aggregates. At date 0, a liquid-income flow $Z > 0$ moves from H to S . At date 1, a public reserve may finance an operating input. Prices and the gross return on that reserve are normalized to

$$P_0 = P_1 = R_B = 1.$$

The effective run configuration closes the baseline household accounts as follows:

$$\begin{aligned} d_{\text{adm}} &= 1, \quad d_I = 1, \quad C_{H0} = 72, \quad C_{S0} = 8, \\ S_{j0} &= Y_{j0} - C_{j0} \quad (j \in \{H, S\}), \\ (S_{H0}, S_{S0}) &= (8, 12). \end{aligned}$$

The spending coefficients are local affine responses over the finite synthetic allocations studied here, not global consumption functions. In particular, initial consumption is not obtained by multiplying initial income by a marginal spending response.

2 Fiscal capture and closed accounts

Let τ be the statutory rate, λ_{legal} legal coverage, and χ_{comp} compliance. Gross receipts are

$$B_{\text{legal}} = \lambda_{\text{legal}} Z, \quad T = \tau \chi_{\text{comp}} B_{\text{legal}} = \kappa_{\text{gross}} Z, \quad \kappa_{\text{gross}} = \tau \lambda_{\text{legal}} \chi_{\text{comp}}. \quad (1)$$

The government pays an administrative cost C_{adm} and allocates the remaining receipts across a transfer y_T , capacity investment I_q , a reserve B_0 , and an unused financial balance U_0 :

$$T = C_{\text{adm}} + y_T + I_q + B_0 + U_0, \quad (2)$$

$$R_B B_0 = X_1 + B_1. \quad (3)$$

All fiscal allocations are nonnegative:

$$C_{\text{adm}}, y_T, I_q, B_0, U_0, X_1, B_1 \geq 0.$$

The reserve is a public deposit asset with an equal depository liability. It is not date-0 spending. Administrative and investment purchases enter the declared first-round measure once; the later spending of their suppliers is outside that measure.

Household income changes are

$$Y_H = Y_{H0} - Z + y_T, \quad Y_S = Y_{S0} + Z - T.$$

With $m_T = m_H$, consumption and saving obey

$$C_H = C_{H0} + m_H(-Z + y_T), \quad S_H = S_{H0} + (1 - m_H)(-Z + y_T), \quad (4)$$

$$C_S = C_{S0} + m_S(Z - T), \quad S_S = S_{S0} + (1 - m_S)(Z - T). \quad (5)$$

End-of-date liquid assets satisfy $A_{j1} = A_{j0} + S_j$. The implementation checks nonnegative consumption and the applicable liquid-asset restrictions for every reported allocation.

3 The two gates

3.1 Direct demand

Let d_{adm} and d_I denote the portions of administrative and investment purchases included in first-round domestic expenditure. The direct demand change is

$$\Delta D_0^{\text{dir}} = -m_H Z + m_S(Z - T) + m_H y_T + d_{\text{adm}} C_{\text{adm}} + d_I I_q. \quad (6)$$

The demand gate passes when $\Delta D_0^{\text{dir}} \geq 0$. This is an incidence result under fixed prices and fixed spending responses. An output or inflation result would require an additional equilibrium closure.

3.2 Capacity and opportunity

Inherited capacity depreciates and may suffer damage:

$$q_1^{\text{pre}} = (1 - \delta_q)q_0 - d_q, \quad (7)$$

$$q_1 = \Pi_{[0,1]}(q_1^{\text{pre}} + \phi I_q). \quad (8)$$

Operating input and capacity are complements:

$$\mathcal{S}_1^O = \min\{aX_1, bq_1\}, \quad \mathcal{O}_1 = \mathcal{O}_0 + \rho_O \mathcal{S}_1^O. \quad (9)$$

The opportunity gate passes when $\mathcal{O}_1 \geq \mathcal{O}_{\min}$.

Define the service requirement

$$s_{\text{req}} = \frac{[\mathcal{O}_{\min} - \mathcal{O}_0]_+}{\rho_O}.$$

The minimum-input formulas maintain the primitive conditions $a, b, \rho_O, \phi > 0$. For $0 < s_{\text{req}} \leq b$, the componentwise minima are

$$X_{\min} = \frac{s_{\text{req}}}{a}, \quad q_{\min} = \frac{s_{\text{req}}}{b}, \quad I_{\min}(\phi) = \frac{[q_{\min} - q_1^{\text{pre}}]_+}{\phi}, \quad B_{\min} = \frac{X_{\min}}{R_B}. \quad (10)$$

For a fixed transfer, the opportunity target is financeable exactly when

$$y_T + I_{\min}(\phi) + B_{\min} \leq T - C_{\text{adm}}. \quad (11)$$

If $s_{\text{req}} = 0$, the minima are zero. If $s_{\text{req}} > b$, the target is technically impossible because projected capacity cannot exceed one.

4 Synthetic parameter set

The parameters are deliberately synthetic and selected to expose the four gate regions. Their units are part of the computational contract.

Table 1: Synthetic parameters and units

Symbol	Value	Unit	Interpretation
ω_H	0.800	population share	Exposed households
ω_S	0.200	population share	Owners
Y_H	80.000	date-0 goods	Initial aggregate income of H
Y_S	20.000	date-0 goods	Initial aggregate income of S
A_H	10.000	date-0 goods	Initial liquid assets of H
A_S	90.000	date-0 goods	Initial liquid assets of S
$m_H=m_T$	0.900	goods per income unit	Local spending response of H and transfer recipients
m_S	0.300	goods per income unit	Local spending response of owners
Z	20.000	date-0 goods	Conditional liquid-income displacement
$P_0=P_1$	1.000	price index	Normalized prices
R_B	1.000	gross nominal return	Treasury deposit return
q_0	0.100	capacity index [0,1]	Initial institutional capacity
δ_q	0.100	rate	Capacity depreciation
d_q	0.020	capacity index	Capacity damage
ϕ_H	0.250	capacity per investment good	High conversion capacity
ϕ_L	0.080	capacity per investment good	Low conversion capacity
a	1.000	service per operating good	Operating-input productivity
b	5.000	service per capacity unit	Capacity service ceiling
ρ_0	1.000	opportunity per service	Opportunity conversion
O_0	1.000	opportunity index	Initial opportunity
$O_{\min}$	5.000	opportunity index	Declared opportunity floor

5 Cases A–E

Cases A–D hold the exposed-group response at its benchmark value. Case E keeps the complete fiscal allocation of C and changes only the local spending environment. Cases C and D differ only in conversion efficiency.

Table 2: Five calculated cases and their two gate outcomes

Case	m_H	ϕ	T	y_T	I_q	$B_0 = X_1$	ΔD_0^{dir}	O_1	Demand	Opportunity
A	0.90	0.25	0	0	0	0	-12.0	1.00	fail	fail
B	0.90	0.25	20	19	0	0	0.1	1.00	pass	fail
C	0.90	0.25	20	12	3	4	-3.2	5.00	fail	pass
D	0.90	0.08	20	12	3	4	-3.2	2.55	fail	fail
E	0.45	0.25	20	12	3	4	0.4	5.00	pass	pass

Figure 1 uses $x = \Delta D_0^{dir}$ and $y = \mathcal{O}_1 - \mathcal{O}_{\min}$. The vertical coordinate is therefore a distance to the opportunity target, not the level of opportunity itself.

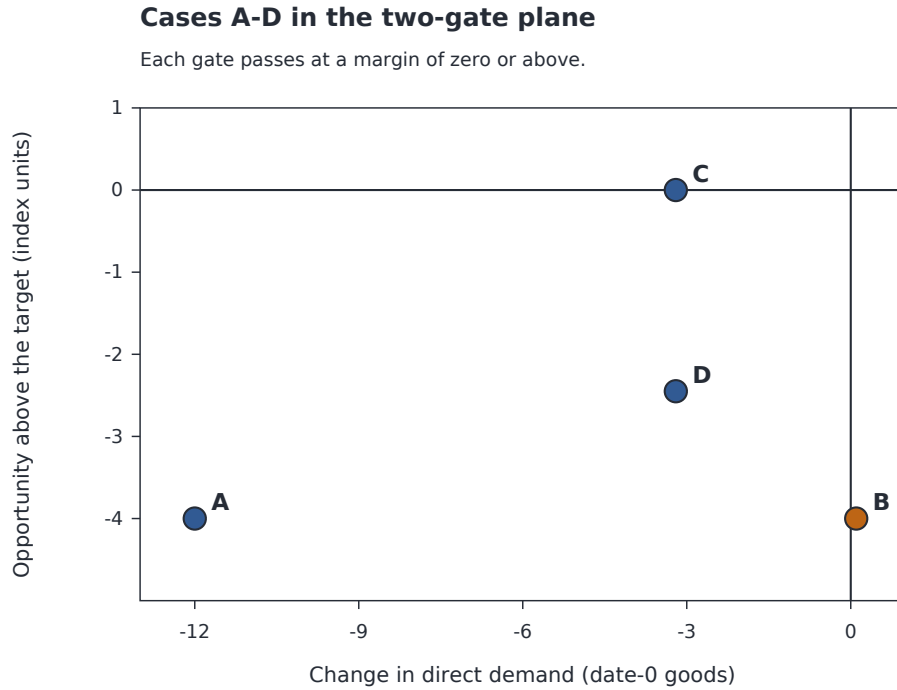


Figure 1: Cases A–D in the two-gate plane. A margin of zero belongs to the passing region.

6 Fixed-allocation frontiers

Holding the fiscal allocation of C fixed gives

$$T = 20, \quad C_{\text{adm}} = 1, \quad y_T = 12, \quad I_q = 3, \quad B_0 = X_1 = 4.$$

The direct-demand expression reduces to

$$\Delta D_0^{\text{dir}} = 4 - 8m_H,$$

so demand passes exactly when

$$m_H \leq m_H^* = \frac{1}{2}. \quad (12)$$

Capacity is $q_1 = \Pi_{[0,1]}(0.07 + 3\phi)$. Because the required capacity is 0.8, the opportunity gate passes exactly when

$$\phi \geq \phi^* = \frac{73}{300} = 0.243\bar{3}. \quad (13)$$

Equality belongs to both passing regions.

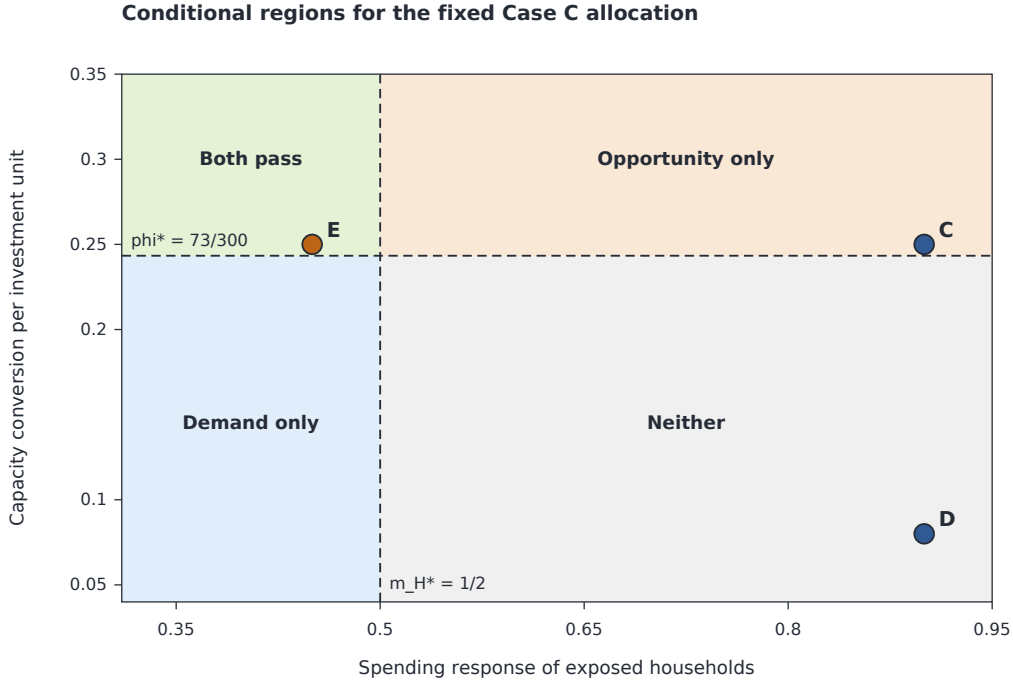


Figure 2: Exact conditional regions under the fixed Case C allocation. The boundaries and the C, D, and E coordinates are read from verified outputs.

7 Minimum finance and boundary cases

Table 3 reports the positive-target calculations for high and low conversion capacity, with and without the fixed transfer. Values are formatted from the unrounded machine-readable fields.

Table 3: Opportunity minima and fiscal feasibility

Capacity	Transfer	ϕ	X_{min}	I_{min}	B_{min}	Total finance	Feasible
High	0	0.250	4.000	2.920	4.000	6.920	yes
High	12	0.250	4.000	2.920	4.000	18.920	yes
Low	0	0.080	4.000	9.125	4.000	13.125	yes
Low	12	0.080	4.000	9.125	4.000	25.125	no

The implementation also checks two cases that a capacity-only shortcut would miss. First, when operation supplies only three service units while four are required, capacity can reach its ceiling and opportunity still remains below the target. Second, when the target is already satisfied, the required investment and reserve are zero even if preinvestment capacity is negative.

8 Allocation under the common fiscal envelope

For a positive and technically attainable opportunity target, reserve the minimum operating balance and set unused funds to zero. Let

$$F = T - C_{\text{adm}} - B_{\text{min}}.$$

When the productive-investment ceiling is imposed, investment is restricted to the amount that can raise capacity to one. Substitution into direct demand gives the affine objective

$$A_D = -m_H Z + m_S(Z - T) + d_{\text{adm}} C_{\text{adm}}, \quad \Delta D_0^{\text{dir}} = A_D + m_H F + (d_I - m_H) I_q.$$

The independent P3 calculation enumerates the complete budget polytope. The four capped vertices and three uncapped vertices establish the two maxima; the numbers below are formatted from the production output, not typed into this document.

Table 4: Demand maxima subject to financing the opportunity floor

Problem	Maximum ΔD_0^{dir}	I_q	y_T	B_0	Opportunity financed
Productive ceiling	-3.128	3.720	11.280	4.000	yes
Ceiling removed	-2.000	15.000	0.000	4.000	yes

The maximum under the productive ceiling is negative. Removing only that ceiling raises the arithmetic maximum but does not make joint passage feasible for the benchmark. The uncapped comparison permits investment labels beyond capacity saturation and is therefore an explicit diagnostic, not a preferred production technology.

9 Accounting verification

The ledger records payer and receiver entries separately and includes the public deposit asset and the depository liability. The residual checks cover individual budgets, the consolidated budget, transaction pairing, fiscal decomposition, capacity, opportunity, domains, and gate classifications.

Table 5: Calculated accounting and domain checks

Case	Checks	Maximum absolute residual	Status
A	70	0.00e+00	PASS
B	70	5.33e-15	PASS
C	70	2.66e-15	PASS
D	70	2.66e-15	PASS
E	70	5.33e-15	PASS

The zero residuals are not treated as evidence about equilibrium prices. They show that the declared incidence experiment and its counterparties close under the stated accounting conventions.

10 What the example establishes

Within its declared closure, the example establishes five operational points.

1. Fiscal capture can be decomposed into rate, coverage, and compliance.
2. Stabilizing current direct demand and producing future opportunity are distinct objectives.
3. A common public envelope may pass one gate, both gates, or neither.
4. Installed capacity cannot replace a missing operating input.
5. The intersection of the two gates is nonempty under the exact conditional frontiers shown above.

The exercise does not solve the parent HANK model, determine the automation income shift, endogenize prices, estimate parameters, or rank policies by social welfare. Those tasks require the additional modules identified in the book.

11 Reproduction contract

The document-build directory contains the LaTeX source, six generated tables, two vector PDF figures, a table manifest, a figure-validation record, and a build report. The required order is:

1. execute a new reproduction run;
2. verify the completed run;
3. execute `build-docs` using that run and a new output directory;
4. inspect every rendered page.

This sequence is autonomous. The package-only `coverage` command checks the structure of the coverage registry without the manuscript. Two optional correspondence checks—`coverage-manuscript` and `check-book`—require an identified copy of manuscript v22 supplied separately. The absence of that protected manuscript is not a failure of the standalone example.

The figures embed Bitstream Vera Sans from the pinned ReportLab dependency. The font is redistributable under the Bitstream Vera Fonts Copyright notice included with the package. Font substitution changes typography and PDF metadata, not the economic coordinates or region definitions.

12 Machine-readable evidence

The run retains unrounded CSV output. The editorial tables apply declared decimal precision only after calculation. The figure-validation record preserves every scenario coordinate, the transformation $\mathcal{O}_1 - \mathcal{O}_{\min}$, both exact frontiers, the grid classification check, units, titles, and axis labels. The book correspondence registry maps each manuscript label to its output file, row key, field, precision, and associated check.